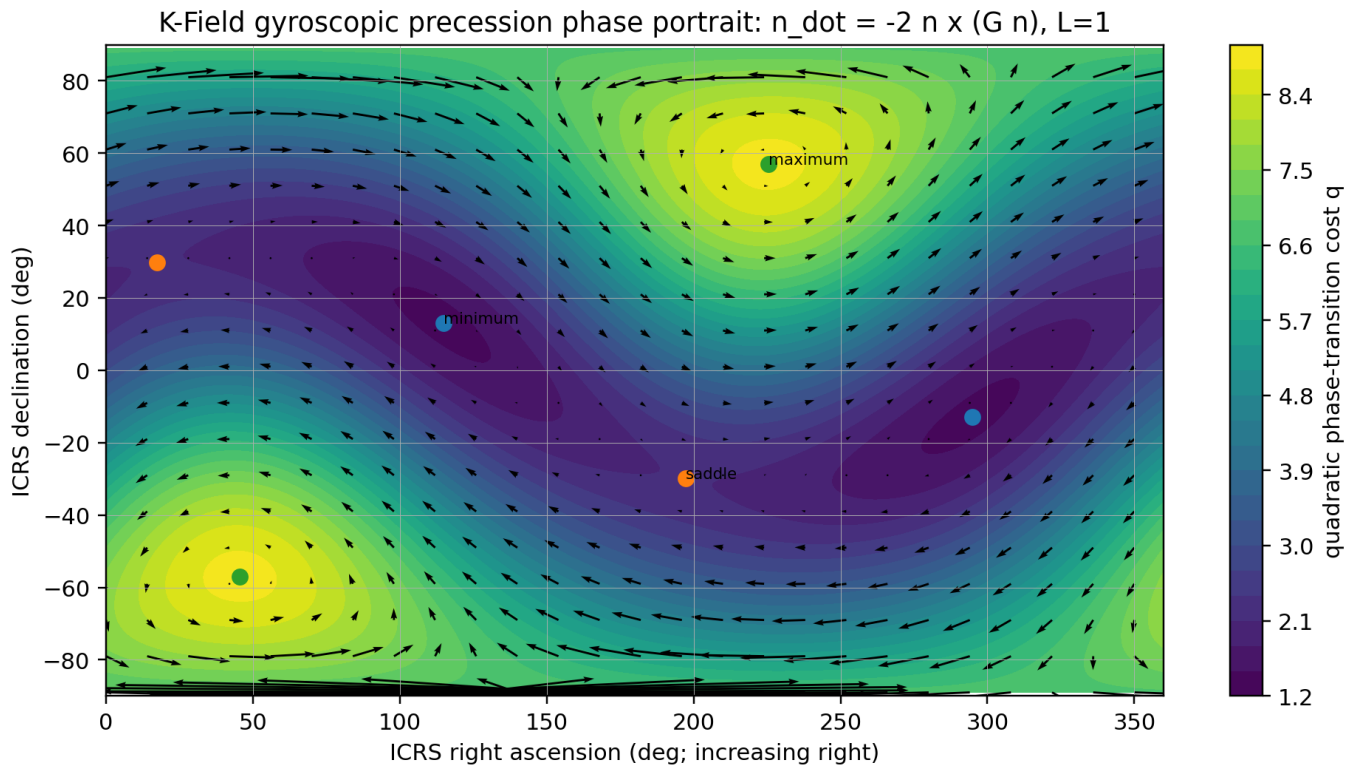


K-FIELD

Galaxy Gyroscopic Precession Geometry

Galaxy axes as distributed gyroscope states, galaxy-plane crossings as collective precession manifolds, and Cell3 as the discrete skeleton of the ensemble phase portrait



Title figure. Exact precession vector field induced by the archived quadratic phase-transition matrix. Arrows show the tangent orientation flow $\dot{n} = -2 n \times (G n)$ for unit spin magnitude; the background is the exact quadratic cost $q(n)$.

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Abstract

This paper establishes a complete mathematical and educational framework for interpreting galaxy orientation statistics as an ensemble of gyroscope proxies moving within the K-Field Quadratic Phase-Transition Cost Surface. A galaxy contributes two linked observables: its projective spin-axis direction and its equatorial plane. The first records a gyroscopic angular-momentum state on the unit sphere. The second generates a great circle whose intersections with other galaxy planes reveal shared orientation manifolds. The exact quadratic matrix, its soft, intermediate, and hard directions, the induced precession equation, the curvature-dependent dwell-time field, the crossing-multiplicity spectrum, the three non-equivalent plane families, and the canonical Cell3 geometry are developed as one internally closed system. The empirical foundation contains exactly 1,000 independent galaxy-group axes, displayed as 2,000 antipodal points but counted projectively once. Galaxy-axis occupancy departs nonlinearly from equal spherical area in the two signed curvature coordinates of the phase-surface slope magnitude. The interaction remains after cost and slope magnitude are controlled. Independently, 499,500 projective plane-pair intersections resolve into 130,553 retained 0.5-degree crossing windows, compared with an isotropic-control mean of 96,840.9 $\pm$ 174.6. The observed high-order tail contains seven windows with at least 20 unique planes and reaches multiplicity 23; the thirty matched controls contain no 22- or 23-plane window. The three recovered plane families exhibit distinct, coherent RA/Dec modulation and harmonic response under stack traversal. The resulting synthesis is direct: galaxy axes are distributed gyroscope states; crossing concentrations are collective precession-plane nodes; curvature occupancy is the dwell-time and focusing record of anisotropic precession; and Cell3 is the discrete geometric skeleton that organizes the ensemble phase portrait.

Foundational Statement

Galaxy axes are distributed gyroscope states, galaxy-plane crossings map shared precession geometry, curvature occupancy records anisotropic dwell time and focusing, and Cell3 supplies the discrete geometric skeleton of the ensemble phase portrait.

Scope, Coordinate Contract, and Data Basis

This reference is internally self-sufficient. It defines the ICRS orientation frame, the projective galaxy-axis population, the quadratic phase-transition matrix, the precession law, the crossing-window statistic, the equal-area curvature estimator, the three recovered plane families, and the canonical Cell3 basis. Numerical tables and figures are generated from the archived matrices and data products listed in the source ledger.

Observed statements, mathematical consequences, and K-Field physical interpretation are kept distinct. Observed statements report catalog counts, fitted orientation trajectories, curvature occupancy, and control comparisons. Mathematical consequences follow from the stated matrices and definitions. Physical interpretation identifies galaxy axes as gyroscope proxies and translates the measured orientation geometry into precession, dwell-time, and collective-manifold language.

1. Galaxies as Gyroscope Proxies and Projective Orientation States

A rotating galaxy carries angular momentum. At the scale used here, the internal details of its stellar, gaseous, and dark components are compressed into an orientation state: the direction of the net spin axis. This compression does not claim that every galaxy is a rigid rotor. It uses the gyroscope as a proxy for the directional persistence of angular momentum. A proxy is sufficient because the observational statistics concern orientation, not the detailed distribution of internal orbital radii.

Let L_i be the angular-momentum vector of galaxy group i and let $n_i = L_i / \|L_i\|$. The observable equatorial plane is perpendicular to n_i . Reversing the axis does not change that plane; n_i and $-n_i$ therefore represent the same projective state. The correct state space is the unit sphere with antipodal identification, written $S^2 / \{+/-1\}$. Figures may display both antipodes to expose the full geometry, but statistical inference counts one representative of each pair.

This separation between display and inference is essential. Counting both antipodes would double every cost and curvature value without adding an independent galaxy. The archived analysis therefore displays 2,000 antipodal points while retaining an inferential population of 1,000 independent projective axes.

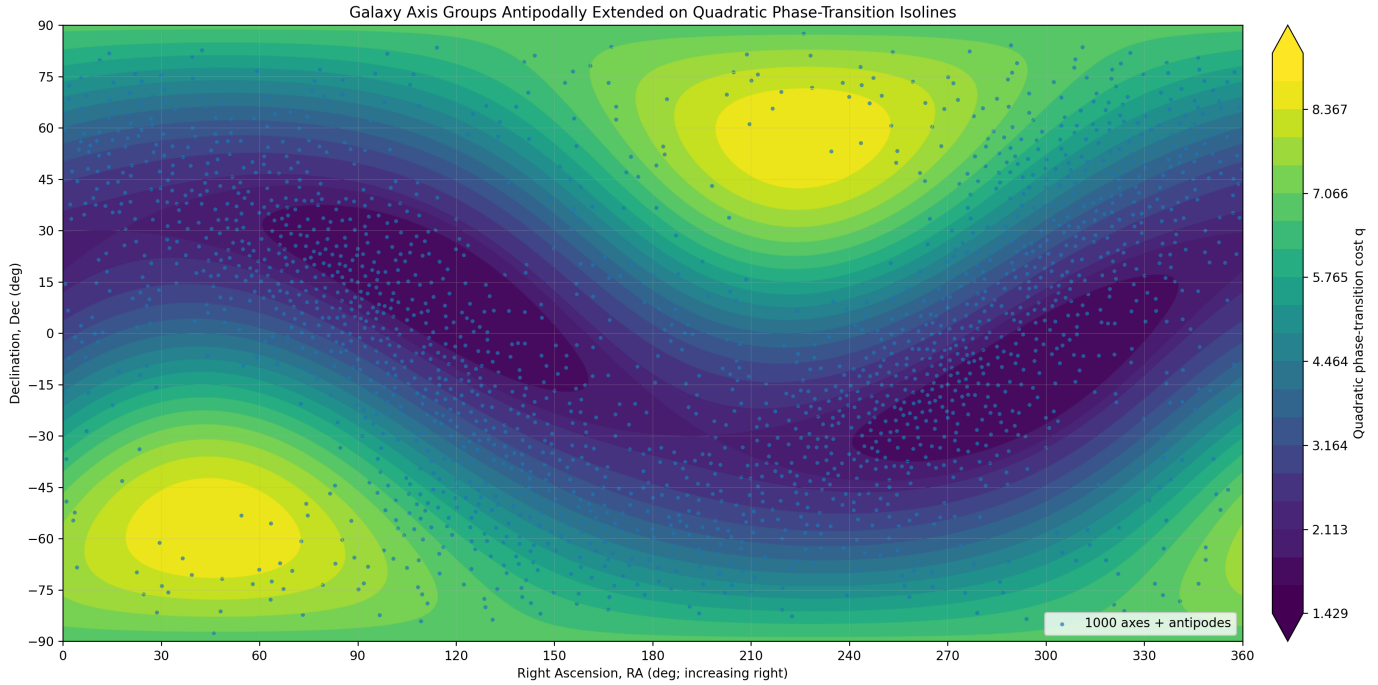
$$\mathbf{n}_i = \mathbf{L}_i / ||\mathbf{L}_i||; \text{ galaxy equatorial plane: } \mathbf{n}_i \cdot \mathbf{x} = 0; \mathbf{n}_i \text{ equivalent to } -\mathbf{n}_i$$


Figure 1.1. The 1,000 independent galaxy-group axes extended to 2,000 antipodal points on the exact quadratic phase-transition cost surface. The display is antipodal; inference counts one projective branch. Source: archived galaxy-axis numerical analysis.

2. The K-Field Quadratic Phase-Transition Cost Surface

The K-Field directional energy geometry is represented by the symmetric quadratic form $q(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}$, with $||\mathbf{n}||=1$. The matrix is fixed in the orthonormal ICRS Cartesian frame. Its x axis points to RA=0 deg, Dec=0 deg; its y axis points to RA=90 deg, Dec=0 deg; and its z axis is the north celestial pole. Right ascension increases from 0 to 360 degrees in the figures, left-to-right, while declination increases upward from -90 to +90 degrees.

A symmetric quadratic form has three mutually orthogonal eigenvectors. Restricted to the unit sphere, these are the stationary directions of q . The smallest eigenvalue defines the soft direction, the largest defines the hard direction, and the intermediate eigenvalue defines the saddle direction. Each direction is projective and therefore has an antipodal partner with the same cost.

The eigenvalue spread is the source of anisotropy. Motion that turns the spin axis toward the hard direction traverses a steeper directional cost geometry than motion toward the soft direction. The surface is not a physical ellipsoid drawn in ordinary position space; it is a scalar field on orientation space. Every point represents a possible angular-momentum direction.

2.1 Exact ICRS matrix

row	G_x	G_y	G_z
0	2.976411	1.243151	-2.08036
1	1.243151	2.592229	-2.317884
2	-2.08036	-2.317884	6.773571

$$q(\mathbf{n}) = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}, \quad ||\mathbf{n}|| = 1$$

2.2 Stationary directions

type	branch	RA (deg)	Dec (deg)	q eigenvalue
minimum / soft axis	+ branch	114.813091	12.969892	1.42932891
minimum / soft axis	- branch	294.813091	-12.969892	1.42932891
saddle / intermediate axis	+ branch	197.222221	-29.836691	2.11254544
saddle / intermediate axis	- branch	17.222221	29.836691	2.11254544
maximum / hard axis	+ branch	225.519007	56.919968	8.80033665
maximum / hard axis	- branch	45.519007	-56.919968	8.80033665

Cell3 plane-family normals and quadratic-cost critical axes

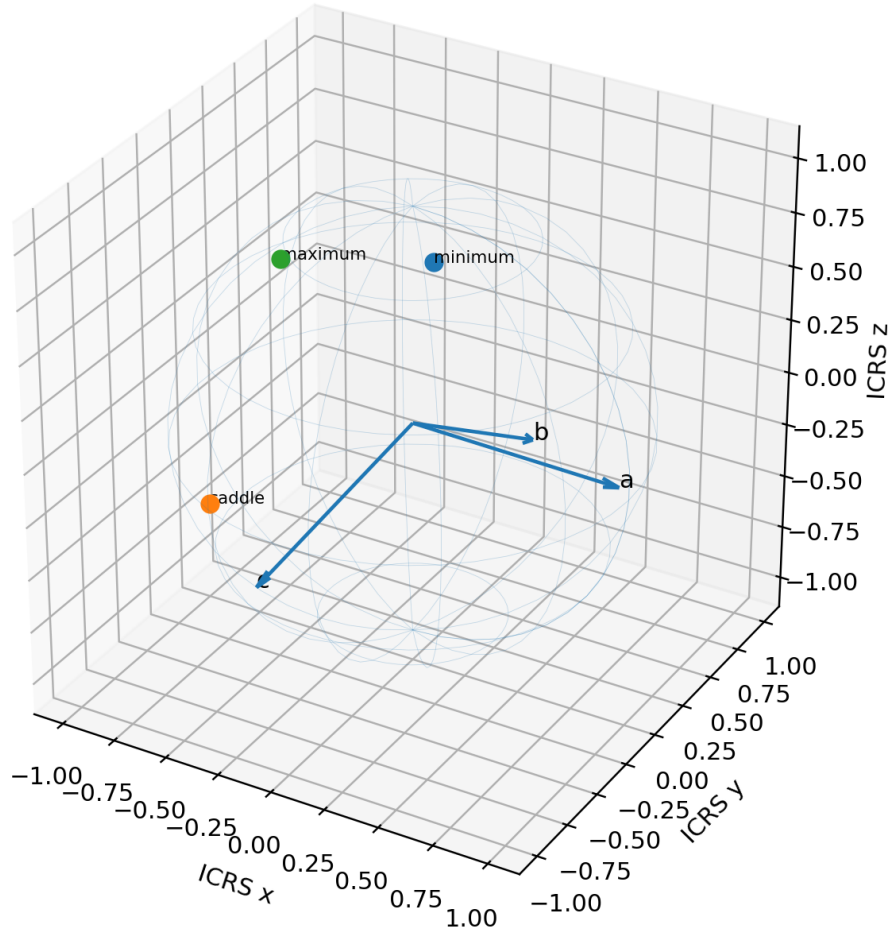


Figure 2.1. The three exact Cell3 plane-family normals and the three quadratic-cost critical directions in the same ICRS Cartesian frame.

3. Gyroscopic Precession on the Quadratic Cost Surface

For fixed spin magnitude L , the directional torque generated by the cost gradient is tangent to the unit sphere. With $C(n) = n^T Q n$ and Q symmetric, the unconstrained gradient is $2Qn$. Its torque about the spin axis is $\tau_K = -n \times 2Qn$. Since $L = L n$, the orientation equation is $\dot{n} = -(2/L)n \times Qn$.

The cross product makes $\dot{n}$ perpendicular to n , so the unit norm is conserved. It also makes the torque perpendicular to Qn in the scalar-product sense needed to conserve the quadratic cost. Direct differentiation gives $dC/dt = 2(Qn) \cdot \dot{n} = -(4/L)(Qn) \cdot [n \times Qn] = 0$. The ideal undamped trajectory therefore remains on one constant-cost contour.

The angular speed is $\|\dot{n}\| = (2/L)\|n \times Qn\|$. It vanishes exactly at the eigenvectors, where Qn is parallel to n . Away from those points the speed varies with both direction and spin magnitude. A high-angular-momentum galaxy traverses the same geometric orbit more slowly than a low-angular-momentum proxy under the same normalized field, while the orbit shape is determined by Q .

This flow is the orientation-space analog of free precession in an anisotropic body. The spin axis does not move radially through the scalar surface; it moves tangentially along a contour. The direction of motion is a ninety-degree rotation of the local gradient within the tangent plane. Cost gradient determines contour spacing, while the cross product converts that gradient into precession.

$$\begin{aligned}\tau_K &= -\mathbf{n} \times \text{grad}(C) = -2 \mathbf{n} \times \mathbf{Qn} \\ d(L \mathbf{n})/dt &= \tau_K; \mathbf{n}_{\text{dot}} = -(2/L) \mathbf{n} \times \mathbf{Qn} \\ dC/dt &= 2(\mathbf{Qn}) \cdot \mathbf{n}_{\text{dot}} = -(4/L)(\mathbf{Qn}) \cdot [\mathbf{n} \times \mathbf{Qn}] = 0 \\ ||\mathbf{n}_{\text{dot}}|| &= (2/L) ||\mathbf{n} \times \mathbf{Qn}||\end{aligned}$$

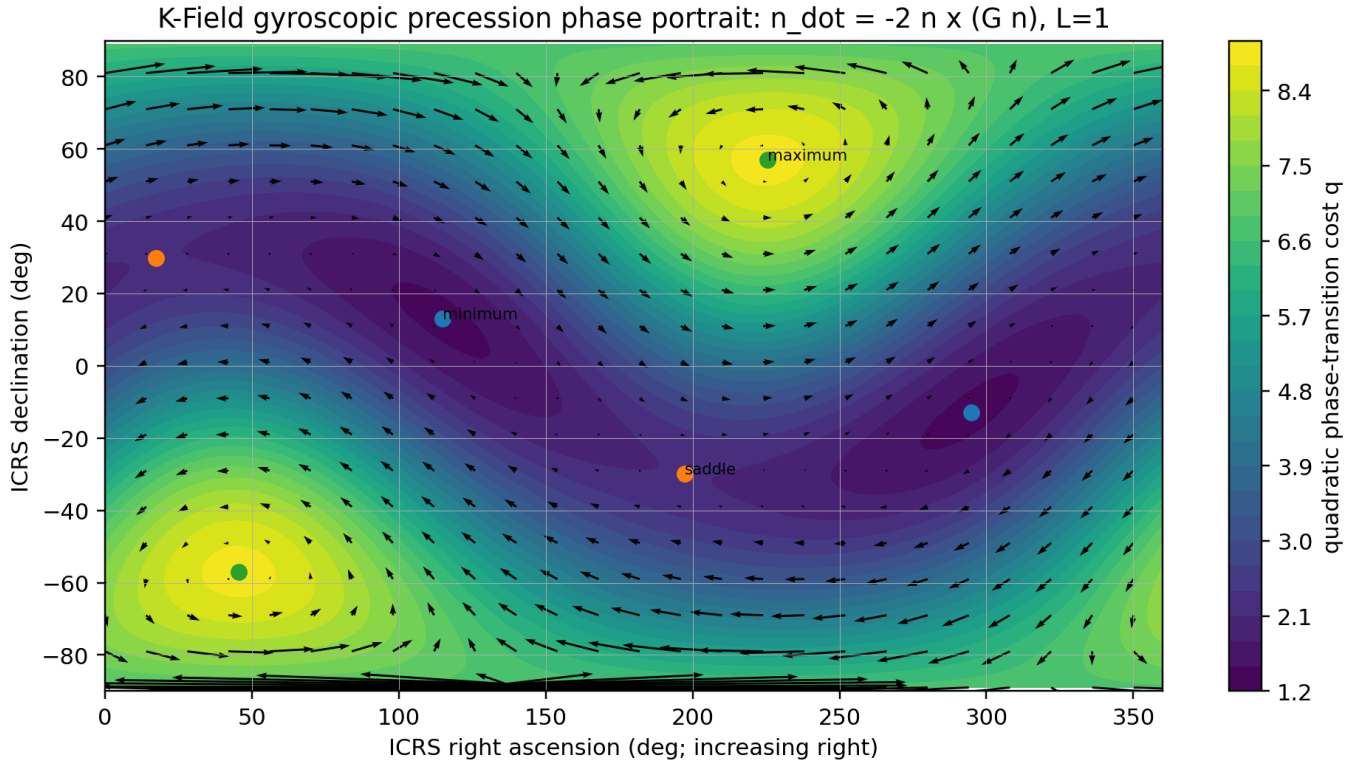


Figure 3.1. Exact anisotropic precession phase portrait. The flow follows constant-cost contours and stops at the three eigen-directions and their antipodes.

4. From One Gyroscope to an Ensemble Phase Portrait

A catalog is an instantaneous ensemble sample, not a time series of one galaxy. The conversion from single-gyro dynamics to population statistics is therefore made through dwell time and phase-space transport. Along a periodic orbit, the fraction of time spent in a small arc is inversely proportional to angular speed. Slow sectors receive greater snapshot occupancy than fast sectors when phases are otherwise mixed.

The quadratic cost value alone does not determine this dwell-time structure. Two positions on the same isoline can have different tangent curvature, different local turning rate, and different convergence of neighboring trajectories. The first derivative gives the local slope magnitude. The second derivative gives the local bending and focusing geometry. That is why the archived occupancy analysis evaluates two signed curvature coordinates rather than reducing the sky to cost minima and maxima.

The local Hessian divides the slope-magnitude surface into concave, saddle-like, and convex regions. This sign topology is useful, but it discards magnitude and principal direction. The stronger result is the complete two-axis curvature state, because the occupancy response depends on the joint value of the two signed components.

4.1 Dwell-time weighting

$$\text{snapshot density along one mixed orbit is proportional to } dt/ds = 1 / ||\mathbf{n}_{\text{dot}}||$$

This relation supplies the first-order population bridge. It does not require every galaxy to have the same spin magnitude or the same contour. It requires only that a persistent orientation flow changes residence time and local transport density. The empirical curvature analysis then tests the stronger statement: neighboring trajectories are organized by second-derivative geometry as well as speed.

4.2 Local tangent Hessian

Let $M(\alpha, \delta)$ be the magnitude of the RA/Dec chart gradient of q . The archived analysis differentiates M twice to obtain C_{RA} , C_{Dec} , and the mixed derivative, and separately constructs the covariant east-north Hessian on the sphere. The two eigenvalues classify local slope-magnitude topology as concave, saddle-like, or convex.

5. Measured Galaxy Occupancy in Curvature State Space

The one-axis decile tests already reject uniform occupancy. Signed RA-chart curvature gives likelihood-ratio $G=22.9474$ with $p=0.00632$. Signed Dec-chart curvature gives $G=34.8533$ with $p=6.324e-5$. The RA response is non-monotonic: the first two deciles are enriched to 1.25, middle deciles are depleted or near unity, the ninth rises to 1.15, and the most-positive decile falls to 0.79. The Dec response begins with strong depletion to 0.59 and 0.76, then rises above unity through several middle and upper deciles.

The joint result is more informative. A full cubic model reduces deviance far beyond an additive cubic model, with Delta $G=79.5962$ for three interaction degrees of freedom and $p=3.747e-17$. Therefore the RA-curvature effect changes with Dec curvature and the Dec-curvature effect changes with RA curvature. The occupancy field cannot be reconstructed by multiplying two one-dimensional curves.

Conditioning separates curvature from simpler scalar descriptions. Adding the curvature coordinates to a slope-magnitude model reduces deviance by 65.0277 for ten degrees of freedom, $p=4.004e-10$. Adding curvature after conditioning jointly on quadratic cost and slope magnitude reduces deviance by 32.9749 for eight degrees of freedom, $p=6.223e-5$. Curvature therefore carries independent organization information.

The covariant east-north analysis corrects the convergence of RA coordinate lines near the poles. East-curvature deciles produce $G=70.8216$ with $p=1.051e-11$, and the nonlinear joint structure persists. The result belongs to the spherical tangent geometry rather than being only a rectangular-chart artifact.

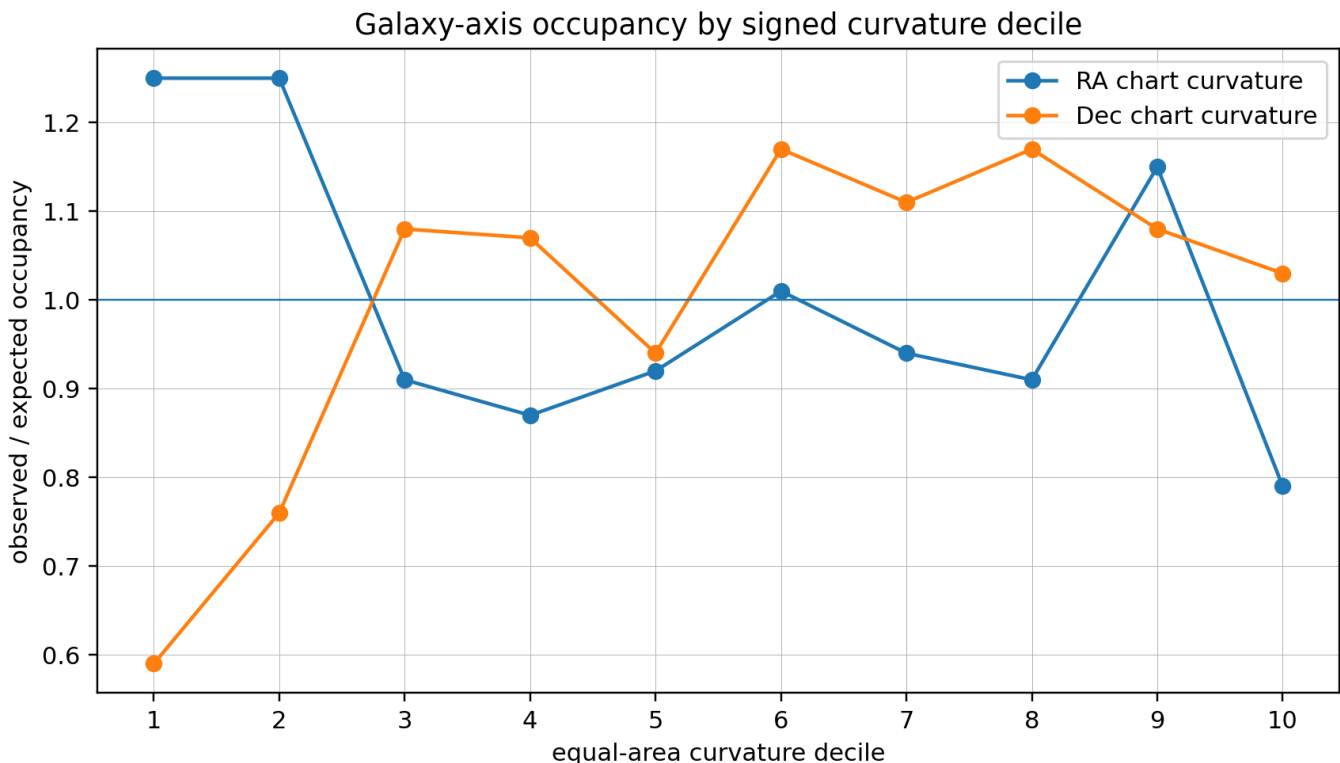


Figure 5.1. Actual observed-to-expected occupancy ratios across the ten equal-area deciles of signed RA and Dec chart curvature.

5.1 Complete RA-curvature deciles

bin	lower	upper	median	observed	expected	ratio	95% low	95% high
1	-inf	-3.934839e-05	-4.266322e-05	125	100	1.25	1.04	1.489
2	-3.934839e-05	-3.340872e-05	-3.630052e-05	125	100	1.25	1.04	1.489
3	-3.340872e-05	-2.792110e-05	-3.057116e-05	91	100	0.91	0.733	1.117
4	-2.792110e-05	-2.181344e-05	-2.499774e-05	87	100	0.87	0.697	1.073
5	-2.181344e-05	-1.557180e-05	-1.892052e-05	92	100	0.92	0.742	1.128
6	-1.557180e-05	-7.821720e-06	-1.207227e-05	101	100	1.01	0.823	1.227
7	-7.821720e-06	2.405453e-06	-2.959675e-06	94	100	0.94	0.76	1.15
8	2.405453e-06	1.565086e-05	8.627023e-06	91	100	0.91	0.733	1.117
9	1.565086e-05	3.271058e-05	2.333049e-05	115	100	1.15	0.949	1.38
10	3.271058e-05	+inf	6.511217e-05	79	100	0.79	0.625	0.985

5.2 Complete Dec-curvature deciles

bin	lower	upper	median	observed	expected	ratio	95% low	95% high
1	-inf	-0.000133	-0.000138	59	100	0.59	0.449	0.761
2	-0.000133	-0.000126	-0.000129	76	100	0.76	0.599	0.951
3	-0.000126	-0.000119	-0.000123	108	100	1.08	0.886	1.304
4	-0.000119	-0.000107	-0.000114	107	100	1.07	0.877	1.293
5	-0.000107	-9.281216e-05	-0.000101	94	100	0.94	0.76	1.15
6	-9.281216e-05	-7.457331e-05	-8.418281e-05	117	100	1.17	0.968	1.402
7	-7.457331e-05	-5.146031e-05	-6.379158e-05	111	100	1.11	0.913	1.337
8	-5.146031e-05	-1.656871e-05	-3.668310e-05	117	100	1.17	0.968	1.402
9	-1.656871e-05	0.000127	2.180767e-05	108	100	1.08	0.886	1.304
10	0.000127	+inf	0.000435	103	100	1.03	0.841	1.249

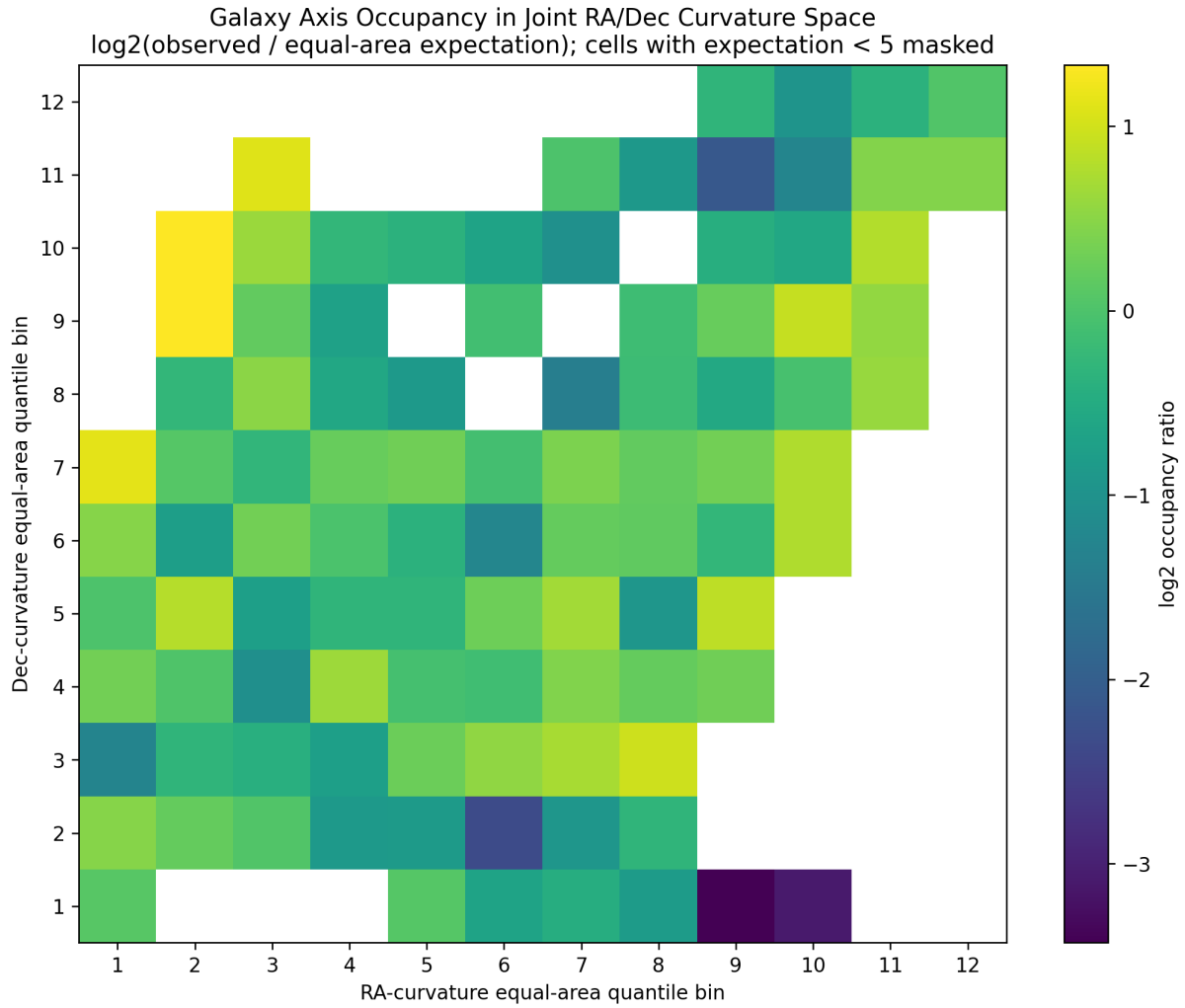


Figure 5.2. Actual joint galaxy occupancy in 12 by 12 equal-area RA-curvature and Dec-curvature quantile space. Color encodes $\log_2(\text{observed}/\text{equal-area expectation})$; low-expectation cells are masked.

5.3 Signed-curvature quadrants

RA sign	Dec sign	area fraction	observed	expected	ratio	95% low	95% high
negative	negative	0.62979167	634	629.791667	1.006682	0.929836	1.088185
negative	positive	0.04875	62	48.75	1.271795	0.975078	1.630384
positive	negative	0.19744599	185	197.445988	0.936965	0.806808	1.082142
positive	positive	0.12401235	119	124.012346	0.959582	0.794934	1.148283

5.4 Nonlinear models

model	degree	interaction	parameters	deviance	residual df	AIC	log likelihood
full polynomial degree 0	0	no	1	332.192877	137	781.102018	-389.551009
full polynomial degree 1	1	no	3	297.148471	135	750.057612	-372.028806
full polynomial degree 2	2	yes	6	288.413466	132	747.322607	-367.661303
full polynomial degree 3	3	yes	10	200.0238	128	666.932941	-323.466471
full polynomial degree 4	4	yes	15	176.613215	123	653.522356	-311.761178
additive polynomial degree 1	1	no	3	297.148471	135	750.057612	-372.028806
additive polynomial degree 2	2	no	5	288.514349	133	745.42349	-367.711745
additive polynomial degree 3	3	no	7	279.620015	131	740.529156	-363.264578
additive polynomial degree 4	4	no	9	277.770131	129	742.679272	-362.339636

The full cubic interaction improves over the additive cubic by Delta G=79.5962 for 3 degrees of freedom, $p=3.747e-17$. The occupancy law is intrinsically two-dimensional.

5.5 Curvature information after conditioning

conditioning design	base deviance	with curvature	base AIC	curvature AIC	Delta G	df	p
M only vs M + curvatures	251.577193	186.549521	556.701312	511.67364	65.027672	10	4.004188e-10
q + M vs q + M + curvatures	276.17797	243.203117	683.789517	666.814664	32.974853	8	6.223412e-05

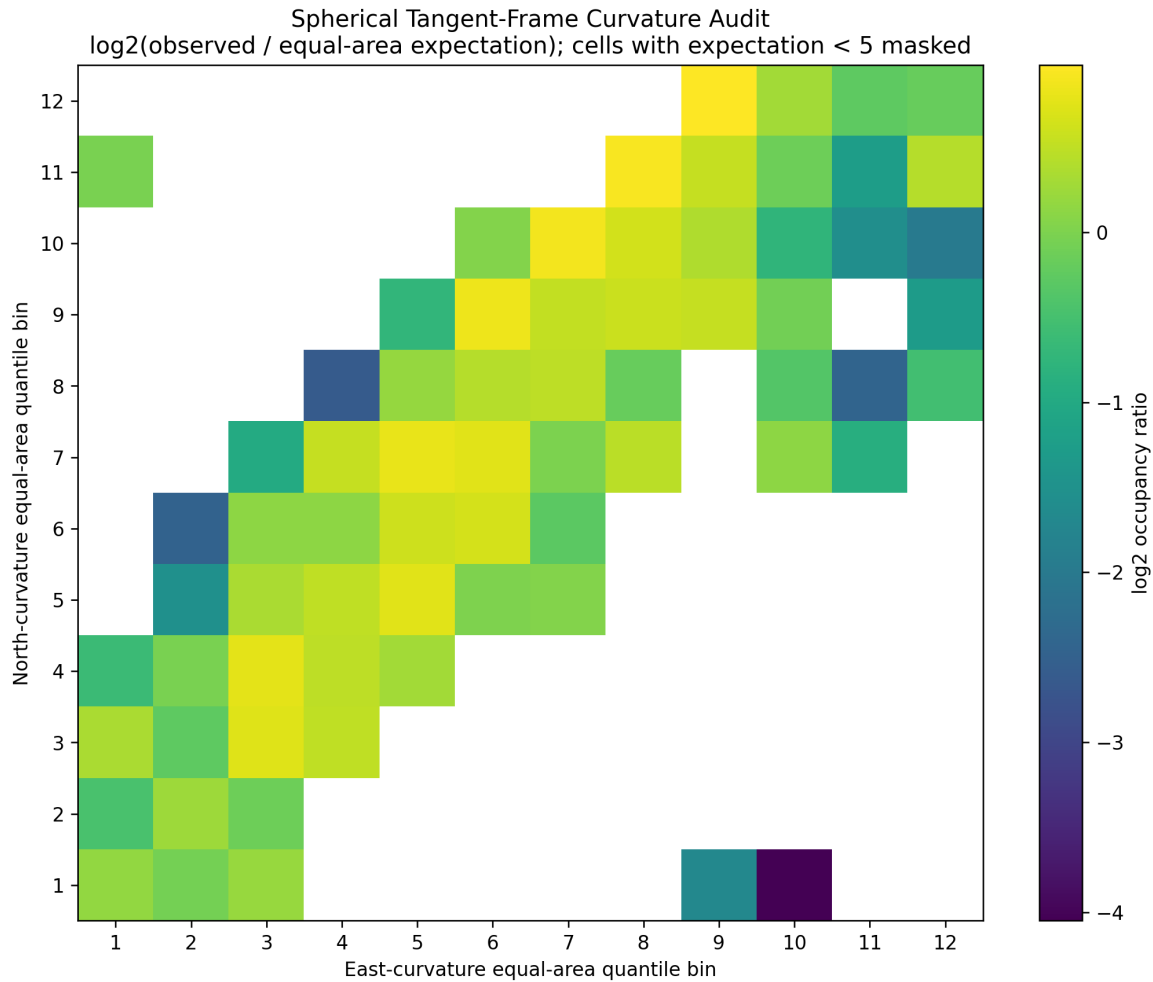


Figure 5.3. Joint occupancy in the covariant east-north curvature frame. The nonlinear structure persists after spherical metric correction.

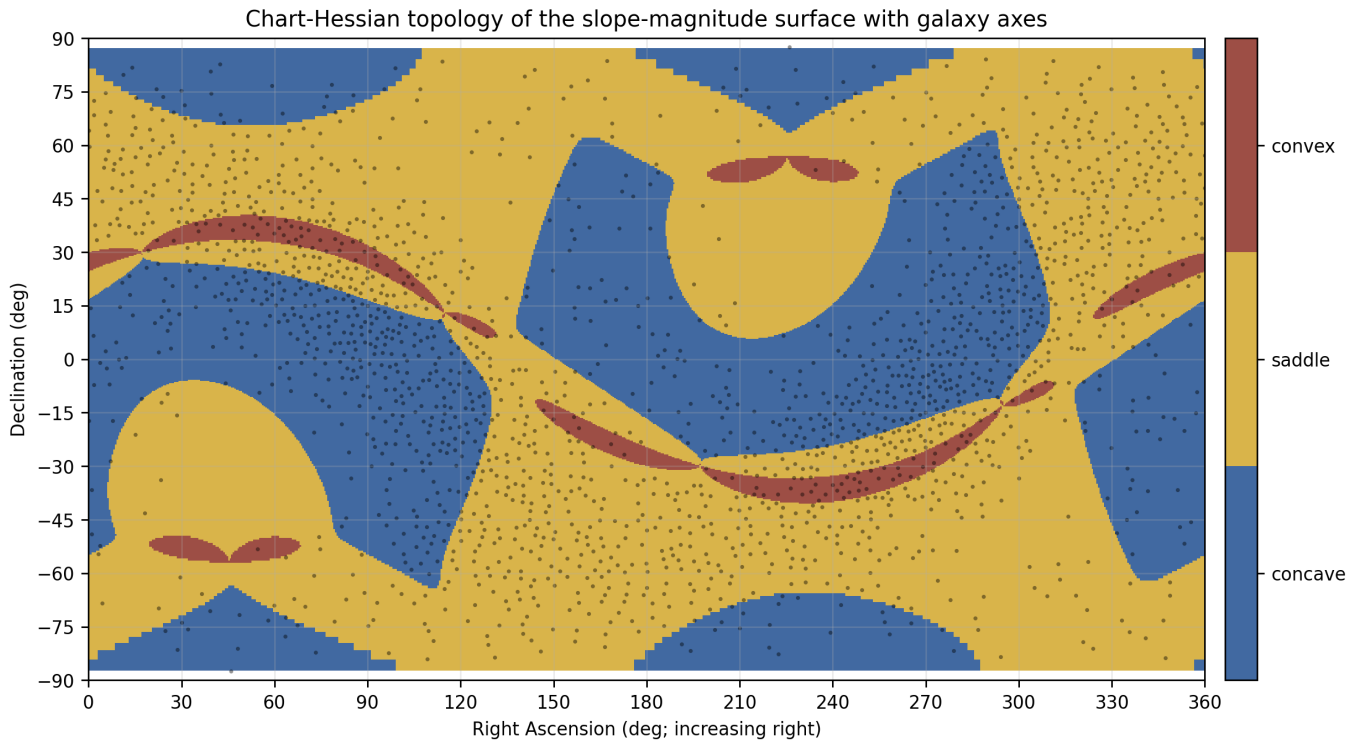


Figure 5.4. Exact chart-Hessian topology with the galaxy-axis population. Concave, saddle-like, and convex regions are defined by the local eigenvalue signs.

frame	topology	area fraction	observed	expected	ratio
RA/Dec chart	concave	0.414606481	409	414.606481	0.986478
RA/Dec chart	saddle	0.541280864	538	541.280864	0.993939
RA/Dec chart	convex	0.044112654	53	44.112654	1.201469
orthonormal tangent	concave	0.526851852	573	526.851852	1.087592
orthonormal tangent	saddle	0.445925926	413	445.925926	0.926163
orthonormal tangent	convex	0.027222222	14	27.222222	0.514286

6. Galaxy Equatorial Planes and Great-Circle Crossing Geometry

Each galaxy equatorial plane passes through the origin. With unit normal $\mathbf{n}_i$, its great circle is the set of unit vectors $\mathbf{x}$ satisfying $\mathbf{n}_i \cdot \mathbf{x} = 0$. Two nonparallel planes intersect along a line through the origin. The two unit-sphere directions on that line are $\mathbf{p}_{ij} = \text{normalize}(\mathbf{n}_i \times \mathbf{n}_j)$ and $-\mathbf{p}_{ij}$.

The point $\mathbf{p}_{ij}$ is not the position of either galaxy. It is the pole of the two-axis orientation plane: both $\mathbf{n}_i$ and $\mathbf{n}_j$ are perpendicular to $\mathbf{p}_{ij}$. If many great-circle crossings fall inside a small angular aperture centered near $\mathbf{p}$, then many galaxy axes lie close to the common great circle $\mathbf{n} \cdot \mathbf{p} = 0$ in dual normal space. High crossing multiplicity is therefore a direct coplanarity statistic for the generating angular-momentum axes.

In gyroscopic language, the crossing center is a collective precession-plane node. The member galaxies need not share one spin axis. They share an orientation manifold: their angular-momentum directions occupy one common plane in projective state space. This distinction explains why the statistic detects structured families without collapsing the population onto a single preferred direction.

```
great circle i:  $\mathbf{n}_i \cdot \mathbf{x} = 0$ 
```

```
 $\mathbf{p}_{ij} = (\mathbf{n}_i \times \mathbf{n}_j) / \|\mathbf{n}_i \times \mathbf{n}_j\|$ ; antipode =  $-\mathbf{p}_{ij}$ 
```

```
high multiplicity near  $\mathbf{p}$  implies  $\mathbf{n}_i \cdot \mathbf{p}$  approximately 0 for many  $i$ 
```

A crossing point is a pole in orientation space, not a galaxy location. It identifies the common plane occupied by many angular-momentum axes.

7. Observed Crossing-Multiplicity Spectrum

The 1,000 observed galaxy planes generate 499,500 unordered projective intersections and 999,000 antipodal surface points. Candidate centers are actual intersections. A geodesic radius of 0.25 degree defines each 0.5-degree-diameter neighborhood. Every nearby pair contributes its two plane identifiers, and the identifiers are deduplicated. The multiplicity is therefore the number of unique participating planes, not the raw number of pairwise points.

Candidates are ordered by unique-plane multiplicity and local pair count. The strongest is retained, and later candidates within 0.25 degree are suppressed. The identical process is applied to thirty complete isotropic sets of 1,000 random plane normals. This control construction preserves the combinatorial dependence among pairwise intersections because the points are generated from full random plane sets rather than randomized independently.

The observed distribution contains 130,553 retained projective windows. The control mean is 96,840.9 $\pm$ 174.6. At multiplicity 14 the galaxy system contains 619 exact windows versus a control mean of 65.7667. At multiplicity 18 it contains 61 versus 0.5. The cumulative tail contains 1,277 observed windows with at least 14 planes versus 98.1333 controls, and 81 with at least 18 planes versus 0.7333 controls.

The highest-order distribution is 20:3, 21:1, 22:2, and 23:1. Seven observed windows reach at least 20 planes. Only three of thirty controls reach 20 or more, two stopping at 20 and one at 21. No control reaches 22 or 23. The observed maximum is 23, while the control maximum mean is 17.63 $\pm$ 1.19 with range 16 through 21.

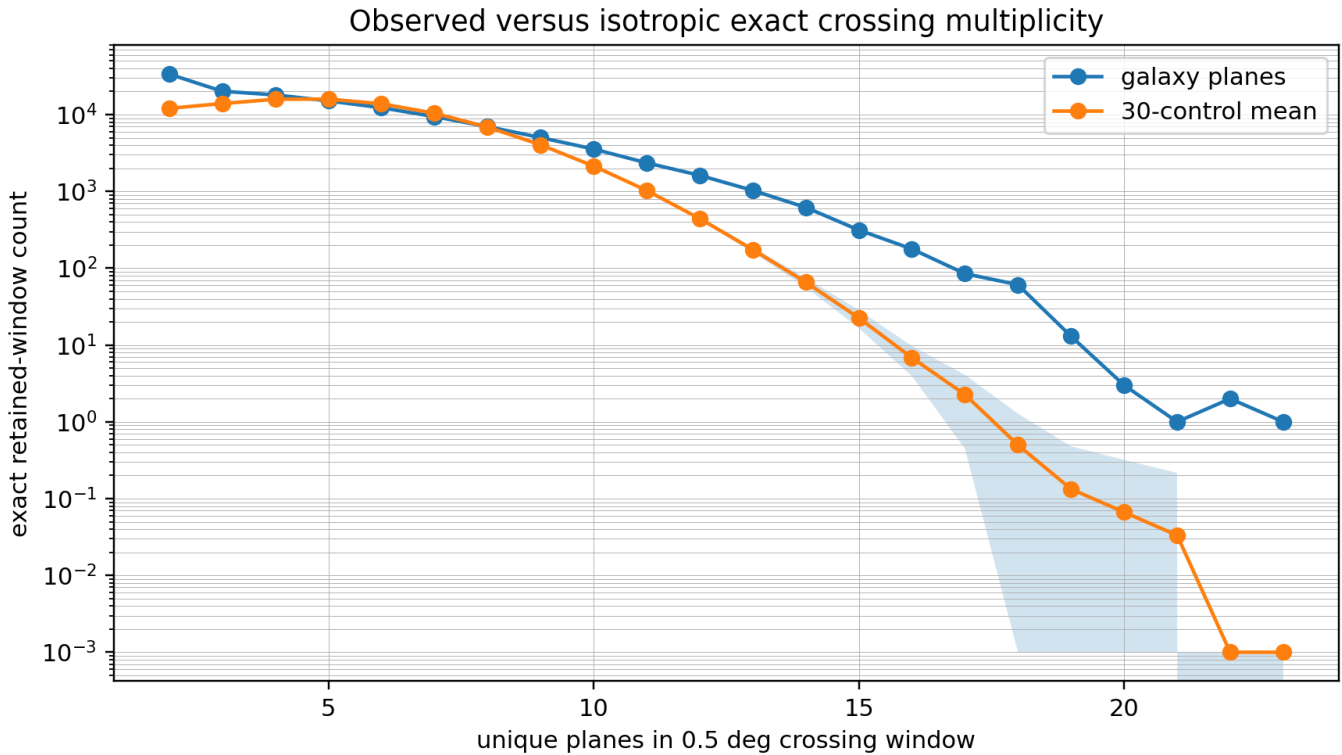


Figure 7.1. Exact retained-window multiplicity for the observed galaxy planes and the thirty-control mean. Counts are logarithmic.

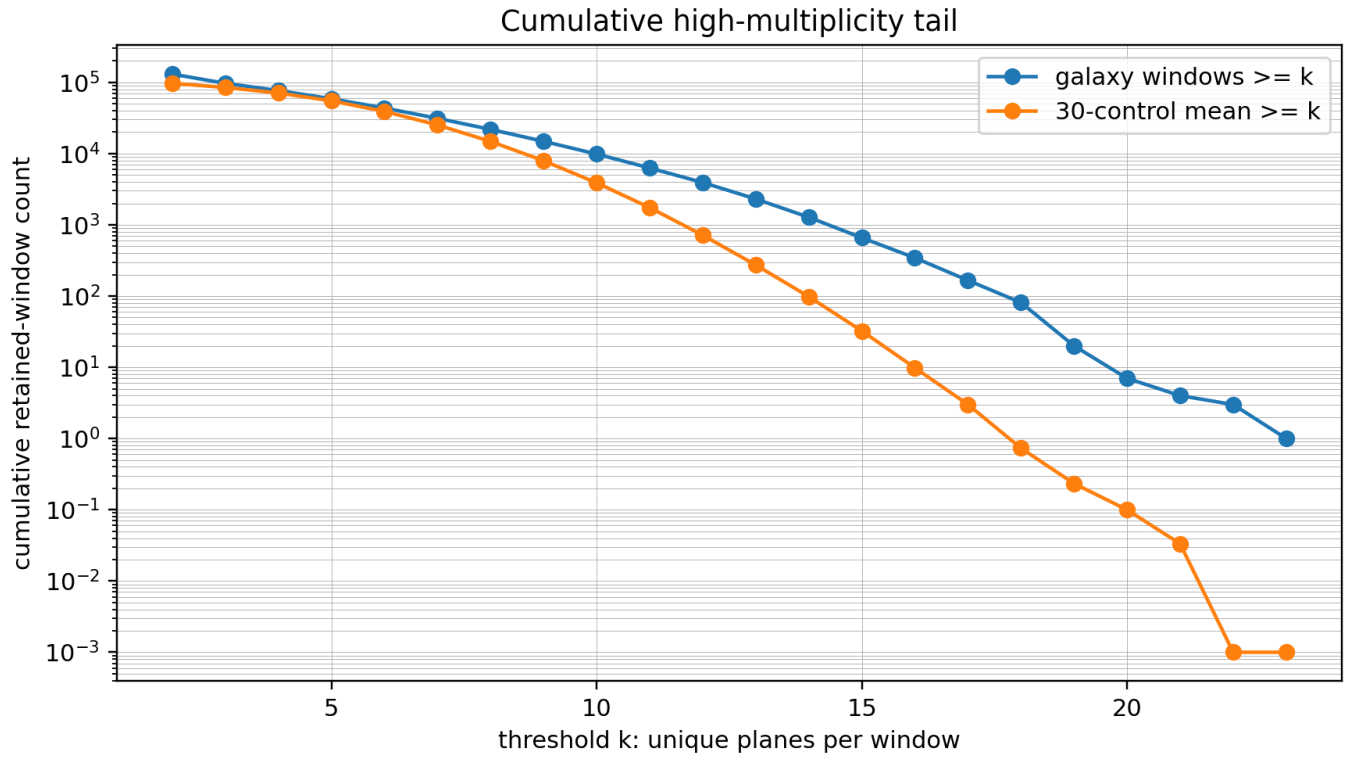


Figure 7.2. Cumulative high-multiplicity tail. The observed separation grows rapidly beyond approximately ten unique planes per window.

7.1 Complete exact multiplicity table

k	observed	random mean	random SD	random 5%	random 95%	observed/random
2	33696	12064.3	100.672	11905	12202.85	2.79303
3	20140	13919.333	138.385	13714.8	14179.3	1.44691
4	18053	15849.433	129.97	15654.6	16017.45	1.13903
5	15102	15936.267	151.968	15783.7	16143.85	0.94765
6	12295	13838.167	121.036	13640.05	14006.95	0.888485
7	9420	10413.733	93.1732	10278.75	10538.7	0.904575
8	6979	6884.467	83.0001	6752.7	7023.25	1.01373
9	5026	4051.5	77.1701	3938.05	4184.6	1.24053
10	3562	2138.733	51.9588	2070.3	2231.6	1.66547
11	2354	1028.667	37.356	969.95	1086.5	2.2884
12	1619	443.333	20.8779	412.45	472.55	3.65188
13	1030	174.833	13.9607	152.45	194.55	5.89133
14	619	65.7667	9.10734	54	79.4	9.41206
15	314	22.5333	5.90577	14	31.55	13.9349
16	178	6.83333	2.87778	1	10.55	26.0488
17	85	2.26667	1.81817	0	6.55	37.5
18	61	0.5	0.776819	0	2	122
19	13	0.133333	0.345746	0	1	97.5
20	3	0.066667	0.253708	0	0.55	45
21	1	0.033333	0.182574	0	0	30
22	2	0	0	0	0	-
23	1	0	0	0	0	-

7.2 Complete cumulative tail

threshold k	observed windows $\geq k$	random mean $\geq k$	random SD $\geq k$
2	130553	96840.9	174.644
3	96857	84776.6	187.713
4	76717	70857.267	164.292
5	58664	55007.833	132.43
6	43562	39071.567	128.024
7	31267	25233.4	123.84
8	21847	14819.667	117.679
9	14868	7935.2	129.322
10	9842	3883.7	90.7103
11	6280	1744.967	66.6351
12	3926	716.3	40.305
13	2307	272.967	24.4096
14	1277	98.1333	14.4287
15	658	32.3667	8.00208
16	344	9.83333	3.80638
17	166	3	2.50517
18	81	0.733333	1.11211
19	20	0.233333	0.568321
20	7	0.1	0.305129
21	4	0.033333	0.182574
22	3	0	0
23	1	0	0

7.3 Thirty matched isotropic controls

replicate	maximum planes in one window	total retained projective windows
1	17	96643
2	18	96865
3	17	97223
4	18	96786
5	17	96837
6	17	96998
7	20	96593
8	16	96969
9	20	97050
10	17	96814
11	18	96598
12	18	96779
13	17	96989
14	17	96752
15	18	97139
16	16	96805
17	18	97040
18	17	96762
19	21	96547
20	19	96882
21	16	96807
22	18	96773
23	17	96624
24	17	96915
25	17	96572
26	17	96938
27	17	96898
28	17	96980
29	19	97010
30	18	96639

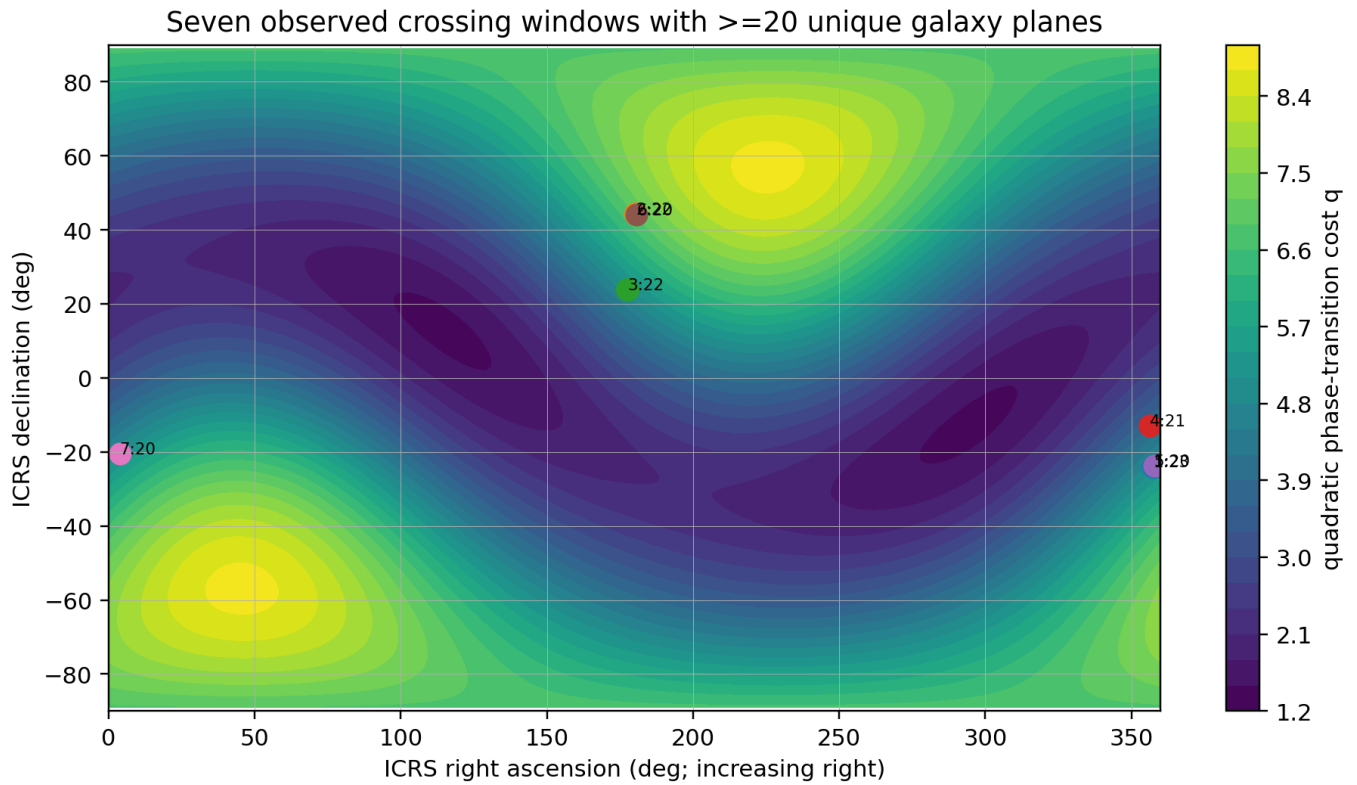


Figure 7.3. The seven observed projective crossing windows with at least 20 unique planes, plotted on the exact quadratic phase-transition cost surface. Labels give cluster ID and multiplicity.

7.4 High-multiplicity membership registry

cluster	RA	Dec	unique planes	local intersections	q cost	plane IDs
1	357.507	-23.9212	23	76	4.9758035	5;9;94;144;163;191;195;201;219;233;243;244;251;344;451;471;481;553;554;568;603;876;967
2	180.475	44.3204	22	64	6.9394063	12;59;85;119;125;200;252;276;285;336;414;484;490;523;570;584;703;845;879;936;978;999
3	177.626	23.9611	22	53	4.9877277	5;9;94;144;163;191;195;201;219;233;244;251;344;451;471;481;553;554;568;603;967;999
4	356.354	-12.8585	21	94	3.8499542	8;92;163;259;292;302;342;360;384;406;424;468;492;581;611;619;747;801;878;973;980
5	357.746	-23.7885	20	75	4.9791092	5;9;144;163;191;195;201;203;219;243;244;251;344;451;471;481;553;568;603;876
6	180.65	44.0919	20	68	6.9348048	12;59;119;125;200;252;276;285;336;414;484;490;523;584;629;703;809;845;978;999
7	3.80398	-20.4292	20	59	5.0406442	2;17;104;141;147;201;221;387;394;399;464;480;503;584;593;601;678;847;861;891

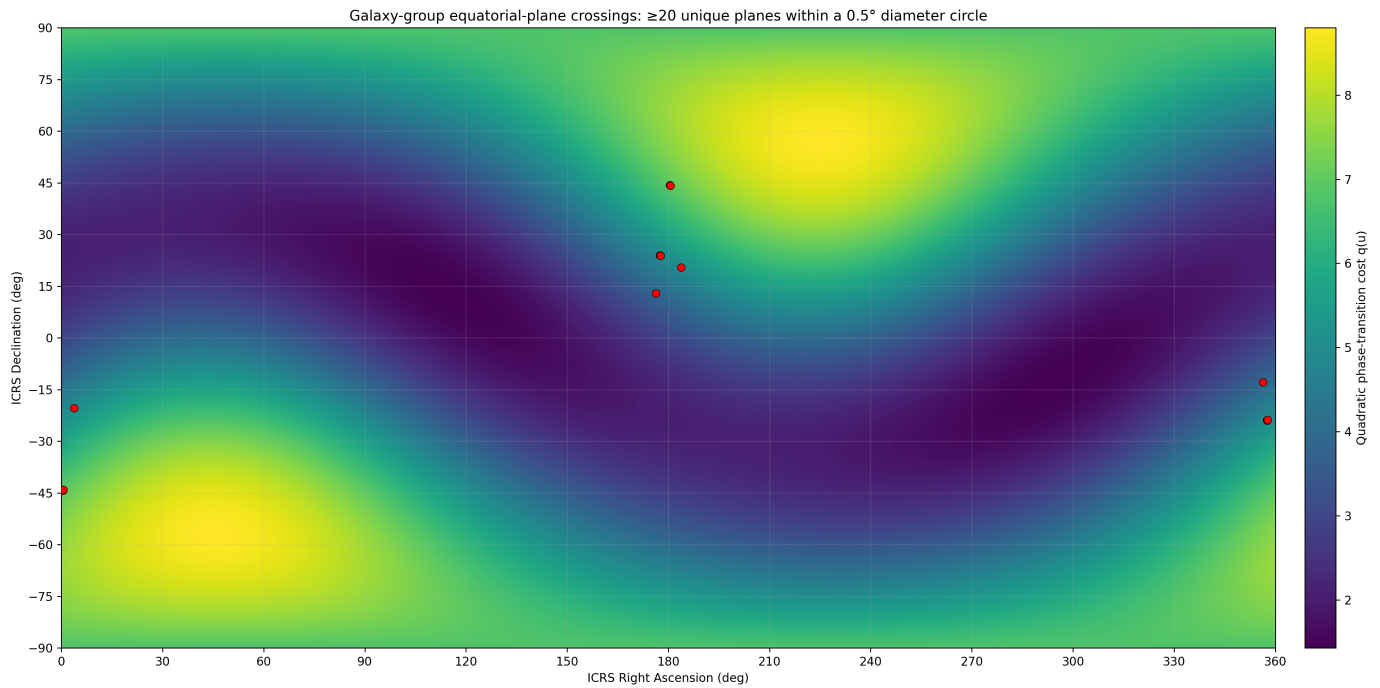


Figure 7.4. Archived actual-data overlay of the ≥ 20 -plane crossing windows on the quadratic phase-transition cost surface.

8. Three Non-Equivalent Gyroscopic Response Families

The three recovered galaxy plane families are not redundant copies. Their mean axis orientations vary coherently through translated stack index k . Family A has an RA peak-to-peak excursion of approximately 28.3 degrees and a Dec excursion of 4.37 degrees. Family B has 14.5 degrees in RA and 5.59 degrees in Dec. Family C has 24.1 degrees in RA and the largest Dec excursion, 9.91 degrees.

Low-order Galactic angular terms capture substantial parts of the family-specific modulation. Family A RA is associated with $\cos(2l)$, correlation 0.698 and $R^2=0.529$. Family B Dec is associated with $\cos(b)$, correlation 0.643 and $R^2=0.653$. Family C RA and Dec each have R^2 near 0.62 under their best simple terms. The harmonic spectra differ by family, and Family C occupies a distinct axis-to-plane-normal regime.

In the gyroscope interpretation, these are three response channels of one anisotropic orientation field. Each family samples a different projection of the precession geometry. Their distinct excursions and harmonics are expected when a non-orthogonal Cell3 basis resolves the field into three independent reciprocal directions.

Galaxy-axis RA, DEC, and phase as functions of translated plane traversal

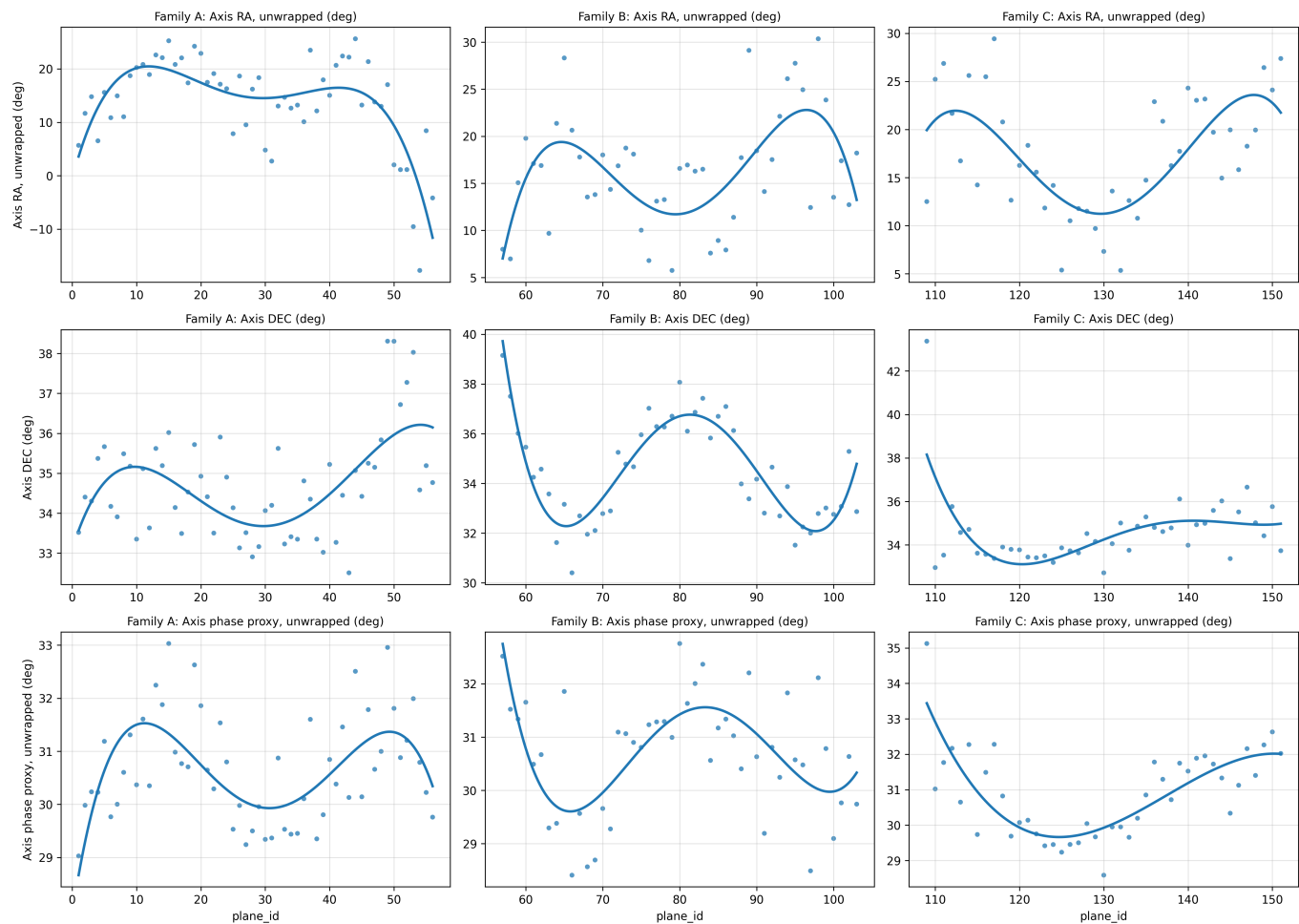


Figure 8.1. Actual plane-level galaxy-axis RA, Dec, and phase trajectories across translated plane traversal for Families A, B, and C.

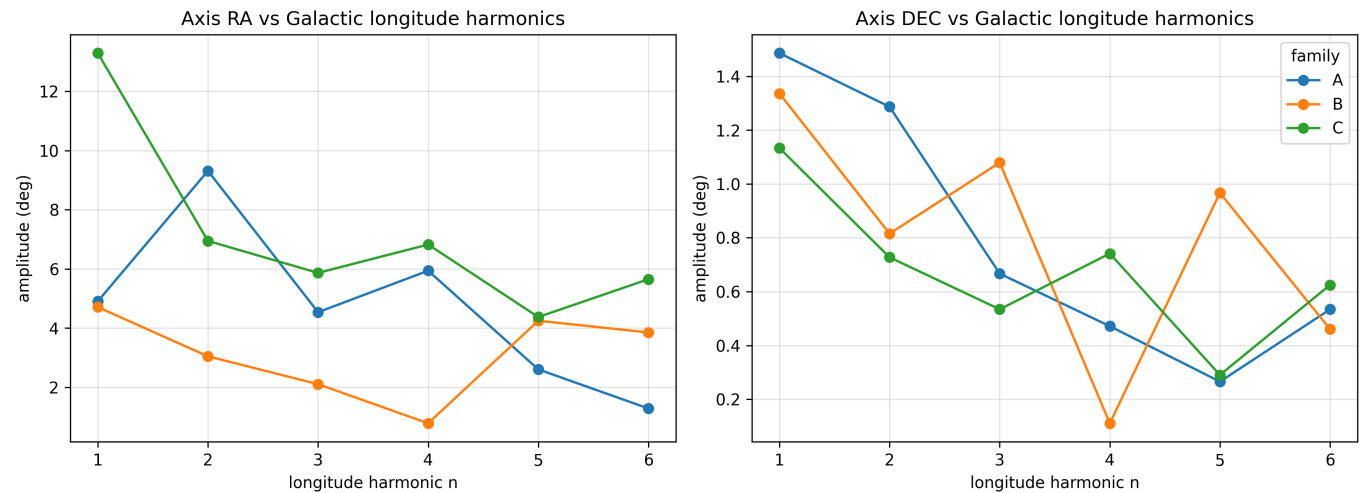


Figure 8.2. Actual low-order harmonic amplitudes of axis RA and Dec versus Galactic longitude. The three families carry distinct harmonic mixtures.

family	measured quantity	value
A	RA peak-to-peak through k	28.3 deg
A	Dec peak-to-peak through k	4.37 deg
B	RA peak-to-peak through k	14.5 deg
B	Dec peak-to-peak through k	5.59 deg

family	measured quantity	value
C	RA peak-to-peak through k	24.1 deg
C	Dec peak-to-peak through k	9.91 deg
A	best simple angular term for RA	cos(2l), corr=0.698, R2=0.529
A	best simple angular term for Dec	sin(l) sin(b), corr=-0.510, R2=0.623
B	best simple angular term for RA	cos(b), corr=-0.455, R2=0.416
B	best simple angular term for Dec	cos(b), corr=0.643, R2=0.653
C	best simple angular term for RA	sin(l), corr=0.540, R2=0.621
C	best simple angular term for Dec	cos(b), corr=-0.533, R2=0.616

9. Canonical Cell3 Geometry: Discrete Skeleton of the Precession Field

The canonical Cell3 basis is $B=N^{-1}D$ in the fixed ICRS frame, with $\det(B)=-1$ and absolute native volume one. The columns a, b, and c are labelled and left-handed. Their lengths are 0.681486335, 1.1958567, and 1.6157628 native units. Their mutual angles are 83.933483, 80.448130, and 50.3632318 degrees. The face areas opposite a, b, and c are 1.488011, 1.08585409, and 0.810396121 native area units.

The 27 Lucas nodes are $p(i,j,k)=B[i,j,k]^T/2$ for i,j,k in $\{-1,0,+1\}$. The constant-sum index $\lambda=i+j+k$ divides them into seven sheets from -3 through +3. The eight nodes with all three intrinsic coordinates equal to ± 1 are the boundary vertices of the centered base cell. Each of the six primitive faces contains exactly nine nodes.

The three plane normals and spacings define the face equations $n_a \cdot x = \pm \delta_a/2$, $n_b \cdot x = \pm \delta_b/2$, and $n_c \cdot x = \pm \delta_c/2$. Since the native volume is one, each opposing face area is $1/\delta$. The Cell3 geometry therefore converts the three independent orientation families into a complete oblique volume with explicit edges, faces, nodes, handedness, and ICRS orientation.

Within the gyro synthesis, Cell3 is not an added drawing. It is the reciprocal-to-primal closure of the plane-family organization. The plane families describe shared angular-momentum manifolds; their intersections define line directions; the three independent families close into the rhombohedral cell. Cell3 is the discrete skeleton of the continuous precession phase portrait.

9.1 Exact basis and invariants

ICRS component	a column	b column	c column
x	0.667867124	0.0720493867	-0.0206565269
y	-0.111950436	0.422783472	-0.794886731
z	-0.0764475597	-1.11630474	-1.40656241
quantity	value	native dimension	
a	0.681486335	length	
b	1.195856702	length	
c	1.615762798	length	
angle(a,b)	83.933483017	degrees	
angle(a,c)	80.448130009	degrees	
angle(b,c)	50.36323175	degrees	
A_bc	1.488011006	area	
A_ac	1.085854093	area	
A_ab	0.810396122	area	
cell volume	1.000000001	volume	
a+b+c	2.740087565	length	

Canonical Cell3 base cell: 8 vertices, 12 edges, and all 27 Lucas nodes

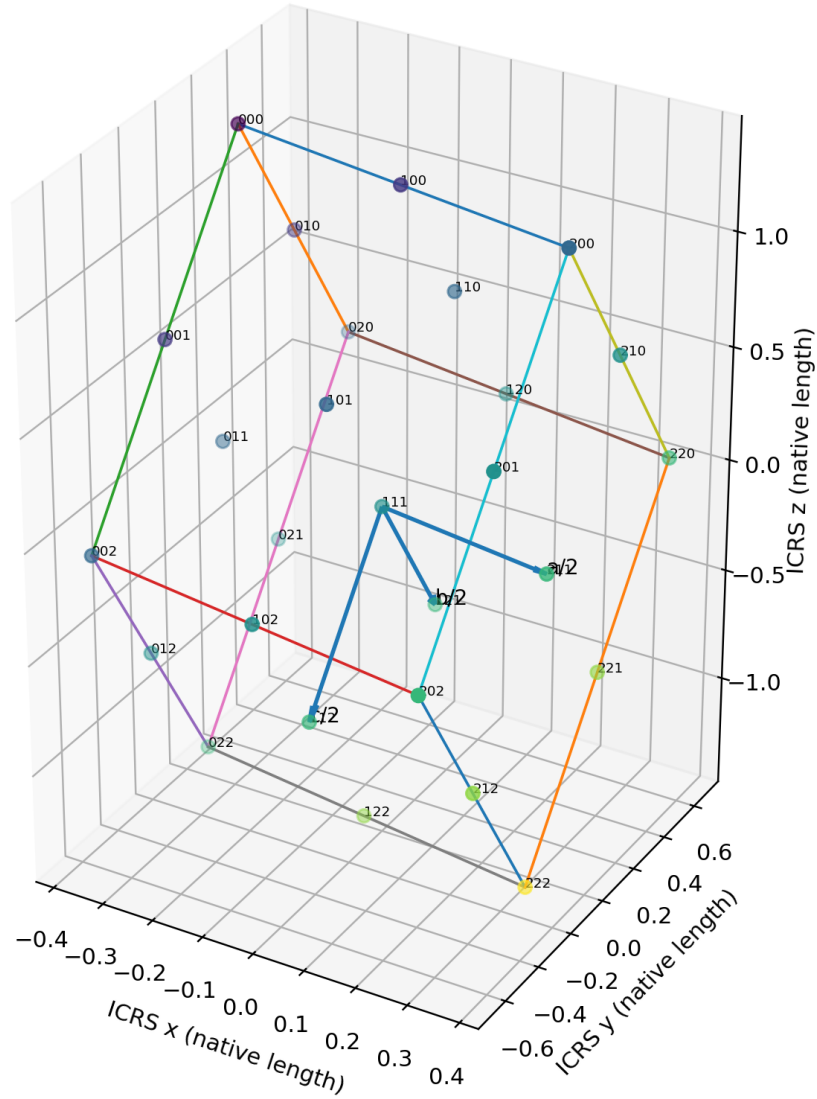


Figure 9.1. Dimensionally correct canonical Cell3 base cell generated directly from B, including all 27 labelled Lucas nodes, all 8 boundary vertices, all 12 cell edges, and the three half-basis arrows.

9.2 Primitive face planes

family	opposite face	n_x	n_y	n_z	RA	Dec	delta	area
a	bc	0.995965	-0.0836022	0.0326193	355.202	1.86928	0.672038	1.488011
b	ac	0.0890526	0.866577	-0.491034	84.1327	-29.4086	0.920934	1.0858541
c	ab	-0.194092	-0.913177	-0.358379	258.001	-21.0007	1.23396	0.8103961

Six centered primitive Cell3 face planes with all 27 nodes

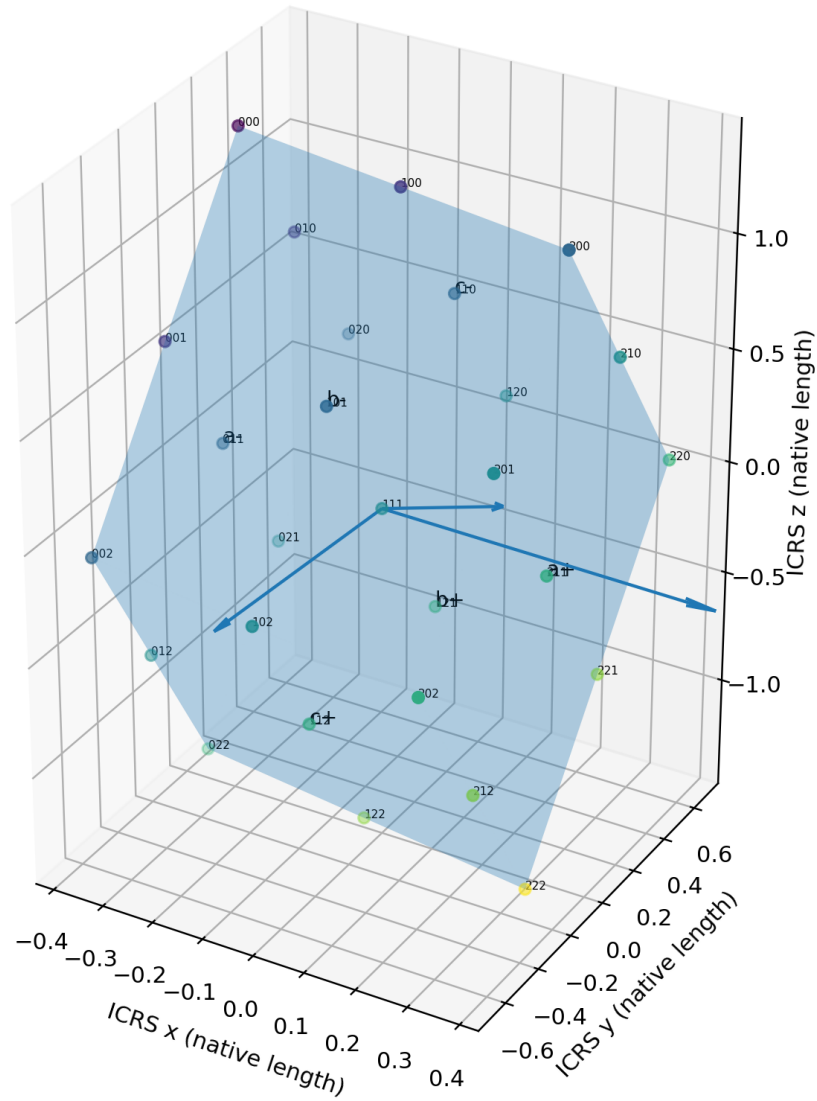


Figure 9.2. Six centered primitive Cell3 face planes with all 27 labelled nodes and the three plane-family normals.

All six primitive Cell3 faces: exact nine-node registries

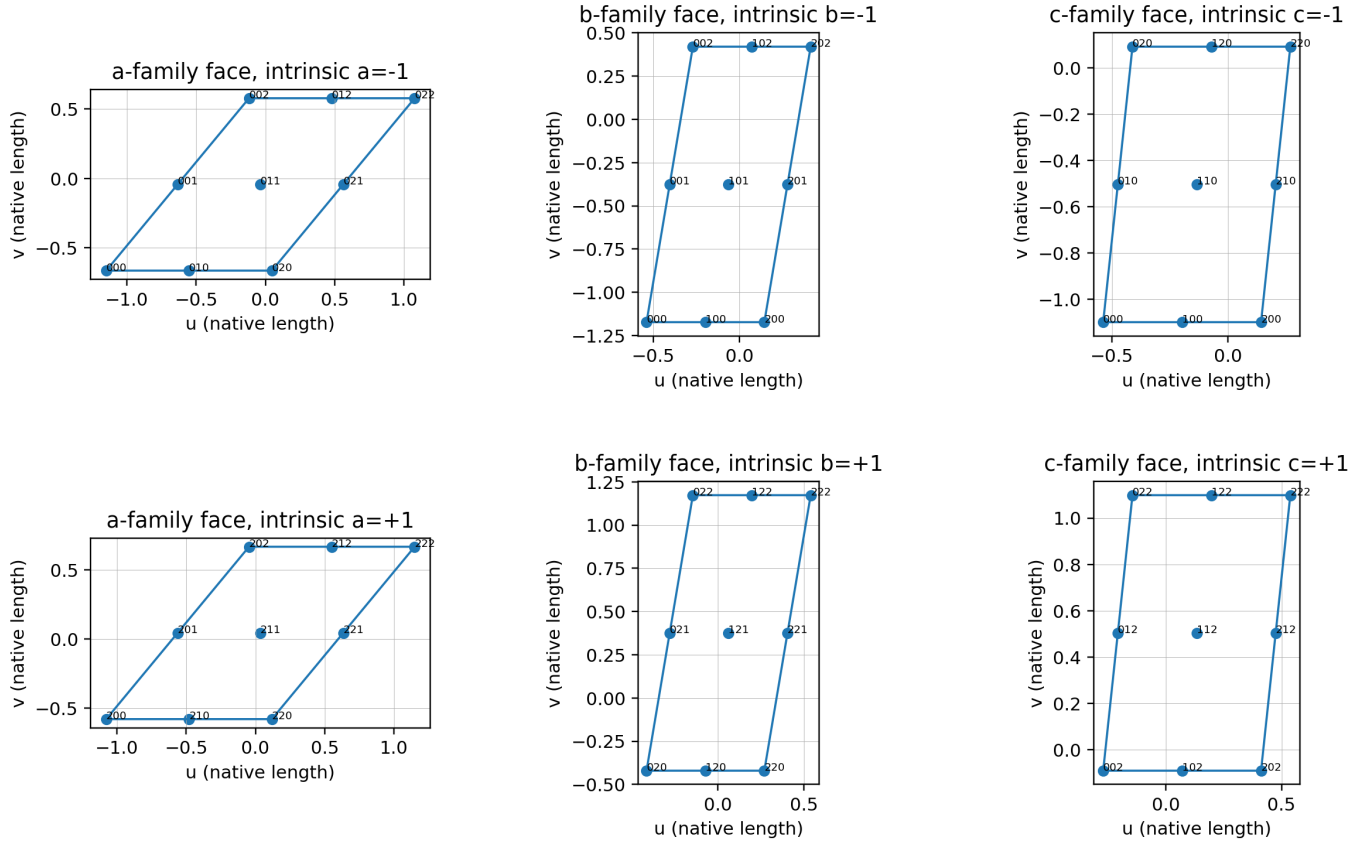


Figure 9.3. All six primitive face panels. Each panel contains the exact nine Lucas nodes on that face, with physical in-plane coordinates in native length units.

9.3 Complete 27-node registry

node	i	j	k	lambda	parity	ICRS x	ICRS y	ICRS z
000	-1	-1	-1	-3	1	-0.359629992	0.242026847	1.299657355
001	-1	-1	0	-2	0	-0.369958255	-0.155416518	0.59637615
002	-1	-1	1	-1	1	-0.380286519	-0.552859884	-0.106905055
010	-1	0	-1	-2	0	-0.323605299	0.453418583	0.741504985
011	-1	0	0	-1	1	-0.333933562	0.055975218	0.03822378
012	-1	0	1	0	0	-0.344261825	-0.341468147	-0.665057425
020	-1	1	-1	-1	1	-0.287580605	0.664810319	0.183352615
021	-1	1	0	0	0	-0.297908869	0.267366954	-0.51992859
022	-1	1	1	1	1	-0.308237132	-0.130076411	-1.223209795
100	0	-1	-1	-2	0	-0.02569643	0.186051629	1.261433575
101	0	-1	0	-1	1	-0.036024693	-0.211391736	0.55815237
102	0	-1	1	0	0	-0.046352957	-0.608835101	-0.145128835
110	0	0	-1	-1	1	0.010328263	0.397443365	0.703281205
111	0	0	0	0	0	0	0	0
112	0	0	1	1	1	-0.010328263	-0.397443365	-0.703281205
120	0	1	-1	0	0	0.046352957	0.608835101	0.145128835
121	0	1	0	1	1	0.036024693	0.211391736	-0.55815237
122	0	1	1	2	0	0.02569643	-0.186051629	-1.261433575
200	1	-1	-1	-1	1	0.308237132	0.130076411	1.223209795
201	1	-1	0	0	0	0.297908869	-0.267366954	0.51992859
202	1	-1	1	1	1	0.287580605	-0.664810319	-0.183352615
210	1	0	-1	0	0	0.344261825	0.341468147	0.665057425
211	1	0	0	1	1	0.333933562	-0.055975218	-0.03822378
212	1	0	1	2	0	0.323605299	-0.453418583	-0.741504985
220	1	1	-1	1	1	0.380286519	0.552859884	0.106905055
221	1	1	0	2	0	0.369958255	0.155416518	-0.59637615
222	1	1	1	3	1	0.359629992	-0.242026847	-1.299657355

10. Canonical Integrated Gyroscopic Interpretation

Then the galaxy-plane statistics become an **ensemble phase portrait of gyroscopic angular momentum** on the K-Field Quadratic Phase-Transition Cost Surface.

For galaxy i , let $\mathbf{n}_i = \mathbf{L}_i / \|\mathbf{L}_i\|$ be its spin axis. Its observed equatorial plane is the rotor plane perpendicular to $\mathbf{n}_i$. Because $\mathbf{n}_i$ and $-\mathbf{n}_i$ define the same plane, the galaxy data naturally occupy projective orientation space.

For the quadratic phase cost $C(\mathbf{n}) = \mathbf{n}^T \mathbf{Q} \mathbf{n}$, the K-Field torque gives $\dot{\mathbf{n}} = -(2/L) \mathbf{n} \times \mathbf{Q} \mathbf{n}$. Therefore $dC/dt = 0$.

An undamped galaxy gyro precesses along a constant-cost contour. The eigenvectors of $\mathbf{Q}$ are stationary spin directions, while the intermediate orientations form anisotropic precession trajectories. The angular speed is $\|\dot{\mathbf{n}}\| = (2/L)\|\mathbf{n} \times \mathbf{Q} \mathbf{n}\|$.

Consequently, a population snapshot is weighted by dwell time: galaxy axes accumulate where precession is slow, dynamically focused, or stabilized by the local tangent Hessian.

This is exactly where the curvature statistics become physically important. The archived analysis shows that galaxy-axis occupancy is a nonlinear function of the two signed local curvature components, and that the relationship remains after controlling for both phase cost and slope magnitude. Thus the galaxies are not merely accumulating at cost minima. Their spin states are organized by the **local geometry of the precession flow** - its focusing, defocusing, saddle structure, and directional stiffness.

The plane-crossing statistic supplies the collective component. For two galaxy gyros, $p_{ij} = (\mathbf{n}_i \times \mathbf{n}_j) / \|\mathbf{n}_i \times \mathbf{n}_j\|$.

A high-multiplicity crossing near $\mathbf{p}$ means that many galaxy axes satisfy $\mathbf{n}_i \cdot \mathbf{p}$ approximately 0.

Therefore, the galaxies are not colocated at p . Their angular-momentum vectors occupy a common orientation plane whose pole is p . In gyro language, the crossing is a **collective precession-plane node**: many independent rotors share the same angular-momentum manifold.

The observed 1,000-plane population contains 130,553 retained crossing windows, compared with 96,840.9 $\pm$ 174.6 for isotropic controls. It has 1,277 windows containing at least 14 planes versus 98.13 randomly, 81 windows containing at least 18 planes versus 0.733 randomly, and seven windows containing at least 20 planes, with a maximum multiplicity of 23. The high-order distribution is 20:3, 21:1, 22:2, 23:1; none of the thirty matched random controls reached 22 or 23.

Interpreted gyroscopically, this heavy tail means that the K-Field does not produce one universal preferred spin axis. It produces a discrete network of preferred **precessional planes, nodal directions, and curvature channels**. The three galaxy plane families then represent three non-equivalent gyroscopic response families. Their axis orientations evolve coherently through stack index, possess different harmonic mixtures, and occupy different axis-to-plane-normal regimes.

The combined interpretation is therefore:

Galaxy axes are distributed gyroscope states, while galaxy-plane crossings map the shared precession geometry.

The individual-axis occupancy measures where gyroscopes dwell on the quadratic cost surface. The crossing multiplicity measures how strongly those gyro states organize into common orientation manifolds. The Cell3 plane families are the discrete geometric skeleton of that ensemble precession field.

11. New Pathways to Discovery and Direct Tests

The combined model supplies a sequence of direct investigations. First, the seven high-multiplicity nodes can be stratified by quadratic cost, slope magnitude, signed curvature, Hessian eigenvectors, and Cell3 address. The result will determine whether the strongest collective plane nodes occupy one derivative-state class or several distinct transport classes.

Second, the analysis can be repeated by galaxy scale, morphology, angular-momentum magnitude proxy, redshift shell, group membership, and environmental density. A genuine gyroscopic transport law predicts systematic changes in dwell-time weighting with L while retaining the same normalized contour geometry.

Third, member-level orientation distributions can replace plane-level summaries. This permits within-family variance decomposition, confidence intervals on each k -index trajectory, and direct estimation of how sharply each galaxy population occupies its assigned precession manifold.

Fourth, the phase-surface vector field can be integrated from every measured axis. The resulting orbit labels, periods, separatrix distances, and local stability exponents provide dynamical coordinates that can be compared with occupancy and crossing multiplicity without reducing the data to RA/Dec bins.

Fifth, the three Cell3 response families can be used as a basis for a tensor transport model. The observed family-specific harmonics then become measured projections of one underlying orientation-flow tensor rather than unrelated sinusoidal fits.

The paradigm change is methodological as well as physical: galaxy orientation is no longer treated only as a position-dependent alignment statistic. It becomes a sampled flow on a directional energy geometry, with measurable conservation law, speed field, curvature field, collective nodes, and discrete reciprocal closure.

12. Exact Reproduction Protocol

Reproduction begins with the 1,000 labelled projective galaxy axes in the fixed ICRS frame and the exact phase-cost matrix. Each axis is normalized, extended antipodally for display, and counted once for inference. Cost, slope magnitude, and signed curvatures are evaluated in 64-bit arithmetic. Equal-area reference samples are uniform in right ascension and $\sin(\text{declination})$, not in rectangular RA/Dec pixels.

For plane statistics, compute every unordered pair cross product, choose one canonical projective branch, and construct a spherical neighbor index at chord radius $2 \sin(0.25 \text{ deg}/2)$. At every actual intersection, pool and deduplicate both plane identifiers from all local pairs. Sort by multiplicity and local pair count, retain the strongest center, and suppress later centers within the same

0.25-degree radius.

For the null model, generate complete sets of 1,000 independent isotropic plane normals and repeat the entire pairwise and suppression pipeline. Do not randomize pairwise intersections independently. For Cell3, use the exact labelled B matrix, generate all 27 nodes from the intrinsic half-step law, and preserve the stated left-handed ICRS embedding without reflection or axis relabelling.

12.1 Computational sequence

1. Load the 1,000 labelled galaxy-group axis directions in the fixed ICRS frame.
2. Normalize each direction, verify unit norm, and create its antipode for display only.
3. Evaluate $q(n)=n^T G_{\text{phase } n}$ and differentiate the slope-magnitude surface in 64-bit arithmetic.
4. Construct a deterministic equal-area reference uniform in RA and sin(Dec).
5. Bin galaxy and reference samples in one- and two-axis curvature quantiles; compute occupancy ratios and likelihood-ratio statistics.
6. Generate all 499,500 projective plane-pair intersections and one canonical branch per antipodal pair.
7. Count unique participating plane identifiers in each 0.25-degree-radius neighborhood and apply matched nonmaximum suppression.
8. Repeat the complete plane construction for each full isotropic 1,000-plane control realization.
9. Generate Cell3 from the exact labelled B matrix and enumerate all 27 half-step nodes and six faces.
10. Integrate the precession vector field $n_{\text{dot}}=-(2/L)n \times G_{\text{phase } n}$ for dynamical continuation.

13. Formula Ledger

concept	formula
ICRS direction	$n=(\cos(\delta)\cos(\alpha), \cos(\delta)\sin(\alpha), \sin(\delta))^T$
projective equivalence	n equivalent to $-n$
quadratic phase cost	$q(n)=n^T G_{\text{phase } n}$
K-Field torque	$\tau_K=-2 n \times G_{\text{phase } n}$
precession equation	$n_{\text{dot}}=-(2/L)n \times G_{\text{phase } n}$
cost conservation	$d q/dt=0$
angular speed	$\ n_{\text{dot}}\ =(2/L)\ n \times G_{\text{phase } n}\ $
galaxy great circle	$n_i \cdot x=0$
pair crossing	$p_{ij}=(n_i \times n_j)/\ n_i \times n_j\ $
crossing coplanarity	$n_i \cdot p$ approximately 0 for participating axes
occupancy ratio	$R=\text{observed/expected}$
likelihood-ratio statistic	$G=2 \sum O \log(O/E)$
Cell3 basis	$B=N^{-1} D; \det(B)=-1$
Lucas node	$p(i,j,k)=B[i,j,k]^T/2$
sheet index	$\lambda=i+j+k$
primitive faces	$n_f \cdot x=\pm \delta_f/2$
unit-volume face area	$A_f=1/\delta_f$

Appendix A. Glossary

term	definition
angular-momentum state	The projective unit direction $n=L/ L $ assigned to one galaxy gyro proxy.
collective precession-plane node	A high-multiplicity great-circle crossing whose pole identifies a common plane containing many galaxy axes.
cost surface	The scalar orientation field $q(n)=n^T G_{\text{phase}} n$ on the unit sphere.
curvature state	The pair of signed local second derivatives of the slope-magnitude surface, evaluated in chart or covariant tangent coordinates.
dwelt-time weighting	The inverse-speed weighting by which a mixed snapshot population spends more time in slow portions of an orbit.
gyroscope proxy	Use of a galaxy spin direction as a persistent angular-momentum orientation without treating the galaxy as a rigid mechanical rotor.
projective sphere	The unit sphere with antipodal directions identified because n and $-n$ define the same galaxy equatorial plane.
response family	One of the three non-equivalent recovered plane stacks that samples a distinct projection of the orientation field.
Cell3	The labelled, left-handed, unit-volume rhombohedral cell recovered from the three plane-family normals and spacings.

Appendix B. Archival Source and Integrity Ledger

source file	bytes	SHA-256
K-Field_galaxy_plane_inersection_distribution_analysis_1784734340.pdf	5,283,885	21bc410dcc8f18c0e733837875665951f7e547ac84cddb738398b8aa2ca740bd
K-Field_galaxy_occupancy_vs_phase_cost_curvature_1784643318.pdf	9,106,337	96c77b991e377bb67d0651463cec76db7a8d9781550f42e6eaa55a84c630dc67
galaxy_axis_radec_plane_stack_subreport.pdf	4,642,748	f7af6054a6372dca13dae66b4fd402d51e18421aebac4bad1b5f120c812d9ed4
K-field_galaxy_sky_rectilinear_projection_analysis_1783781974.pdf	2,630,107	b8bd2763ddd7798f7b364553f46c13f4f6eb8a625da0b3371db127b5a4c7fbc9
K-Field_galaxy_plane_crossings_threshold20_diameter0p5deg_over_phase_cost.png	1,561,864	02c7ee4c0324ed5a3c789b59b038cfcea88771db8ae3fafc79825a0cde034730

The source figures reproduced in this paper are extracted directly from the listed archival PDFs or supplied archival PNG. The newly generated analytical figures are computed from the exact matrices and tabulated observations preserved in the machine-readable companion file.

Conclusion

The galaxy orientation field resolves into one connected physical geometry. Each galaxy axis is a projective angular-momentum state. The quadratic phase-transition matrix defines an anisotropic precession flow with exact conserved cost and direction-dependent angular speed. Curvature occupancy records where the ensemble is focused, depleted, slowed, or redirected within that flow. Great-circle crossing multiplicity detects common planes of angular-momentum states and converts the ensemble into a localized integer-valued node field. The three independent plane families provide non-equivalent response channels whose reciprocal closure is the canonical Cell3 rhombohedral geometry.

Galaxy axes are distributed gyroscope states. Galaxy-plane crossings are collective precession-plane nodes. Curvature occupancy is the derivative-geometric phase portrait. Cell3 is the discrete skeleton of the K-Field gyroscopic orientation field.